

soc lab #001

ENDPOINT HARDENING

DEFENDING THE PERIMETER: A BLUE
TEAM INVESTIGATION INTO END
POINT SECURITY.

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POINT SECURITY ENTHUSIAST.

LAB ENVIRONMENT NOTE: All actions performed in this study were conducted on a dedicated, simulated Windows endpoint. This ensures a "Safety-First" approach to security testing and prevents unintended system downtime.

THE MISSION: ENDPOINT HARDENING

What is Endpoint Hardening? : It is the process of securing a computer system by reducing its Attack Surface. This involves removing unnecessary software, disabling unused services, and most importantly enforcing the Principle of Least Privilege (PoLP).

Why it Matters :

Prevents Persistence: Malware requires administrative rights to stay on a system after a reboot.

Limits Lateral Movement: If one user is compromised, a hardened system prevents the attacker from jumping to other parts of the network.

Mitigates Human Error: It stops users from accidentally installing unauthorized or malicious software.



PHASE 1: IDENTITY SEGREGATION

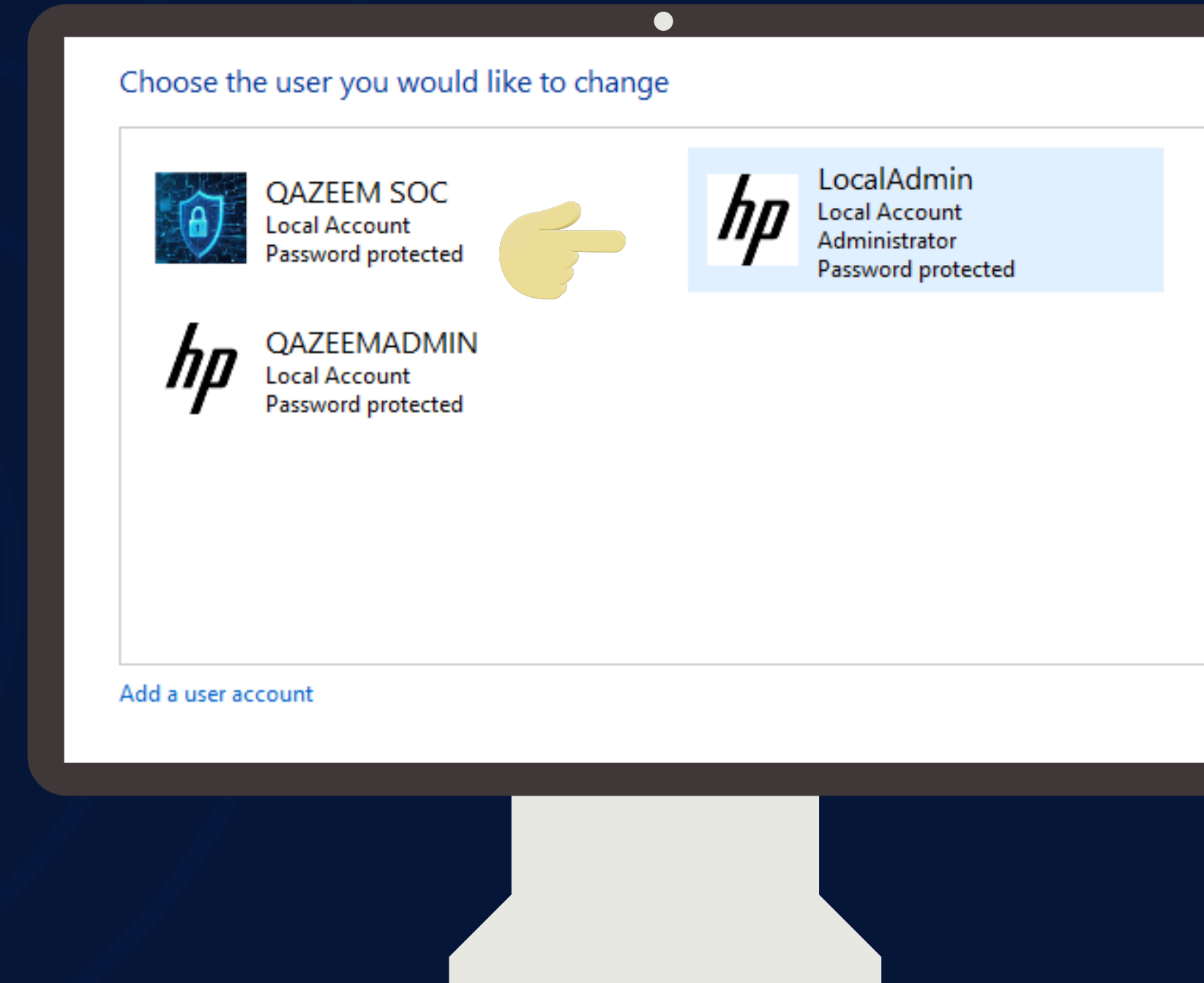
DEFINITION: The practice of splitting a user's digital persona into two distinct accounts to isolate high-risk administrative powers from low-risk daily tasks.

The Action: Establishing a Hardened Administrative Root.

Technical Methodology: Account Creation, Provisioned a standalone, password-protected LocalAdmin account.

Role Assignment: This account is the sole holder of the "Administrative Token," used only for authorized system audits and maintenance.

The Security Result: Daily operations are now demoted from high-privilege credentials. This effectively mitigates Credential Harvesting and Lateral Movement risks.



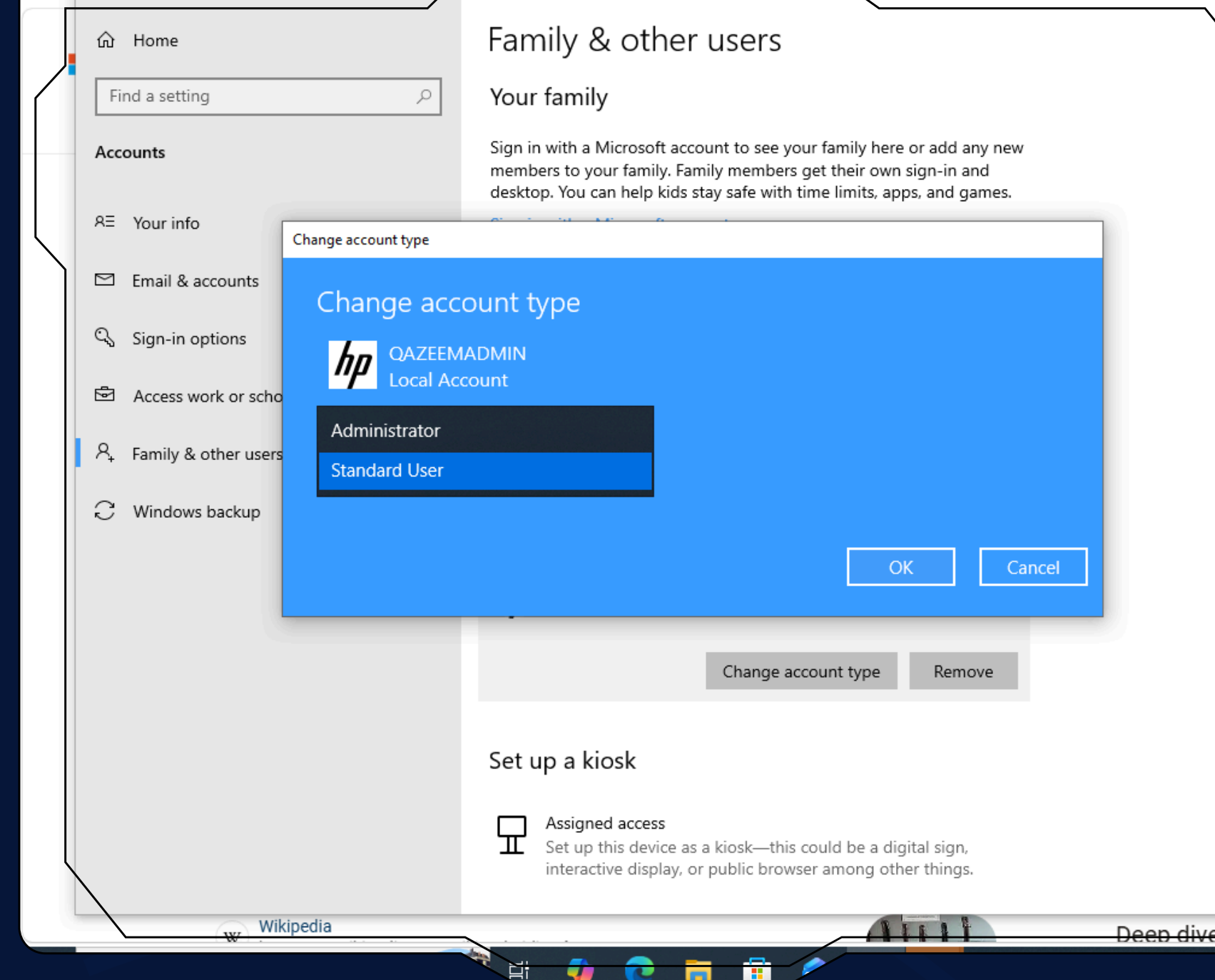
PHASE 2 : PRIVILEGE DEMOTION

DEFINITION: The process of lowering an account's authority level from "Administrator" to "Standard User" to strip away the system-level "Master Key" during daily operations

The Action: Transitioning the Daily User from Administrator to Standard User.

Technical Methodology: Utilized the LocalAdmin root to modify the account type of the primary profile (QAZEEMADMIN).

The Goal: To strip the "Active Administrative Token" from the daily operating environment.



PHASE 3: THE SECURITY AUDIT

DEFINITION: A systematic verification to ensure that security controls are functioning as intended.

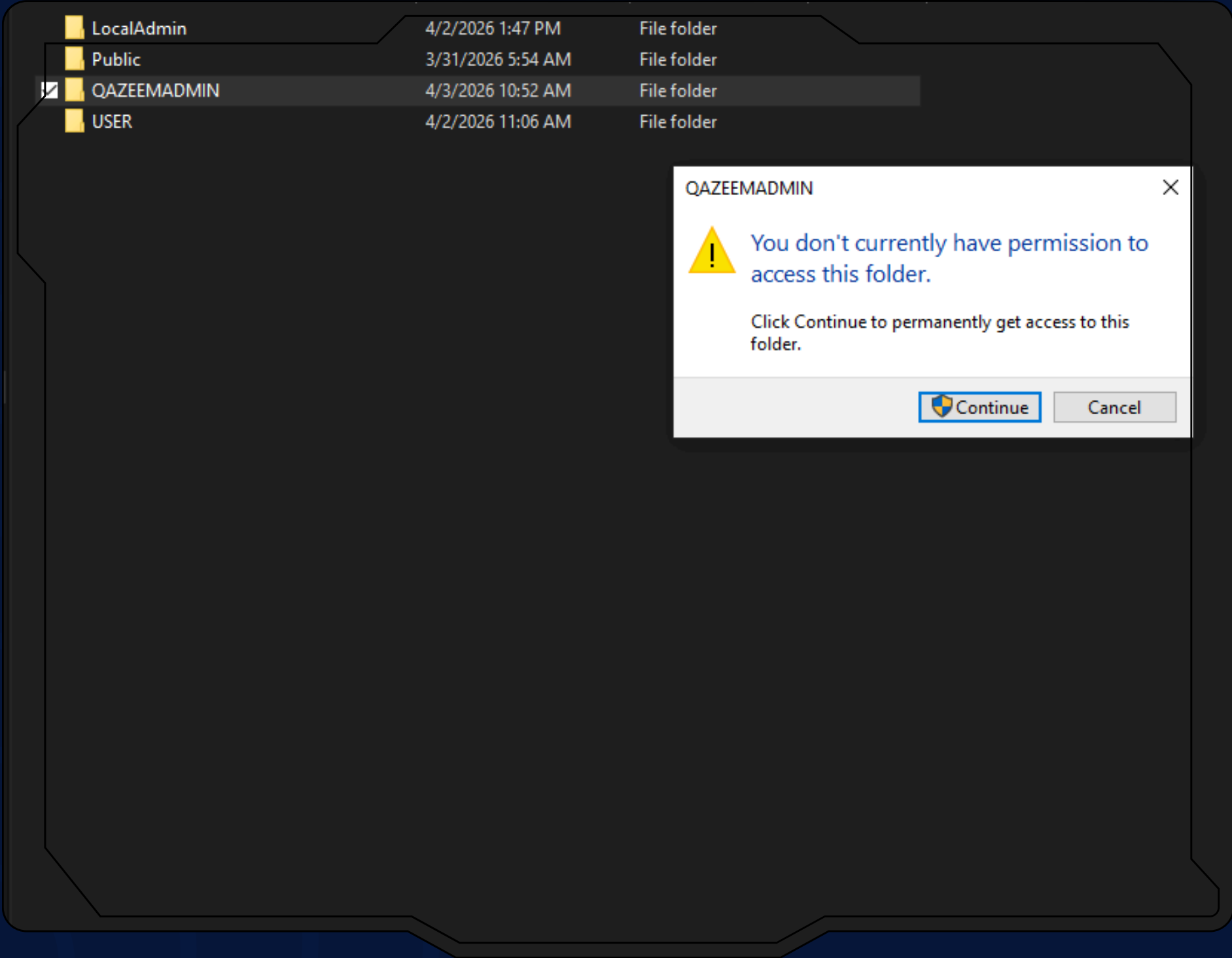
The Test: Attempted unauthorized lateral movement from the Standard User profile into the C:\Users\QAZEEMADMIN directory.

Technical Observation: Interception; Windows File System (NTFS) successfully identified a lack of administrative tokens.

The Challenge: A "Permission Denied" prompt was triggered.

The Barrier: The "Continue" button features the UAC Shield, requiring a password that the daily user does not possess.

Conclusion: **AUDIT PASSED.** The Identity Segregation boundary is active and enforced.



PHASE 4 : SECURITY INSIGHT & ANALYSIS

DEFINITION: The process of evaluating laboratory results to determine their long-term impact on the security posture.

The Analyst's Perspective: Breaking the Chain: By enforcing the Principle of Least Privilege (PoLP), we have effectively broken the "Persistence" chain for 90% of common malware.

Containment: Even if the daily user executes a malicious script, the infection is Sandboxed. It cannot modify the Registry or disable the Firewall because it lacks the Root-level Token.

Strategic Value: We don't just build walls; we build a system where even if the "Front Door" (the user) is compromised, the "Master Safe" (the system) remains locked.



PHASE 5: THE SOC MONITORING PERSPECTIVE



DEFINITION: The continuous observation of system logs to detect unauthorized privilege escalation or account manipulation.

Key Performance Indicators (KPI) for the SOC:

Event ID 4720 (User Creation): Did an attacker create a new "Backdoor" Admin account?

Event ID 4625 (Failed Logon): Is there a Brute-Force attack against the LocalAdmin account?

Event ID 4672 (Special Privileges): Did a standard user account suddenly trigger "Superuser" rights?

The Strategy: Hardening the endpoint is the Shield, but Monitoring the logs is the Eyes. A SOC Analyst must verify that the "Access Denied" blocks we created are actually firing in the logs.

Goal: To reduce the Mean Time to Detect (MTTD) for internal threats.

THE CONCLUSION & KEY TAKEAWAYS

DEFINITION: Summarizing the lab's impact on system integrity and future security architecture.

The Final Verdict: Zero Trust: This lab proves that "Trust" is a vulnerability. By verifying every action via UAC Elevation, we have secured the OS.

Operational Hygiene: Endpoint Hardening is not a one-time task; it is a standard that must be audited regularly.

The Analyst's Growth: Moving from a "User" mindset to a "Defender" mindset requires understanding both the tool and the threat.

The Mission Statement: "I don't just secure systems; I build resilient environments where security is the foundation, not an afterthought."

BY: QAZEEM SAMSHUDEEN TEMITOPE Aspiring SOC Analyst



APPENDIX: PROOF OF WORK & TOOLING

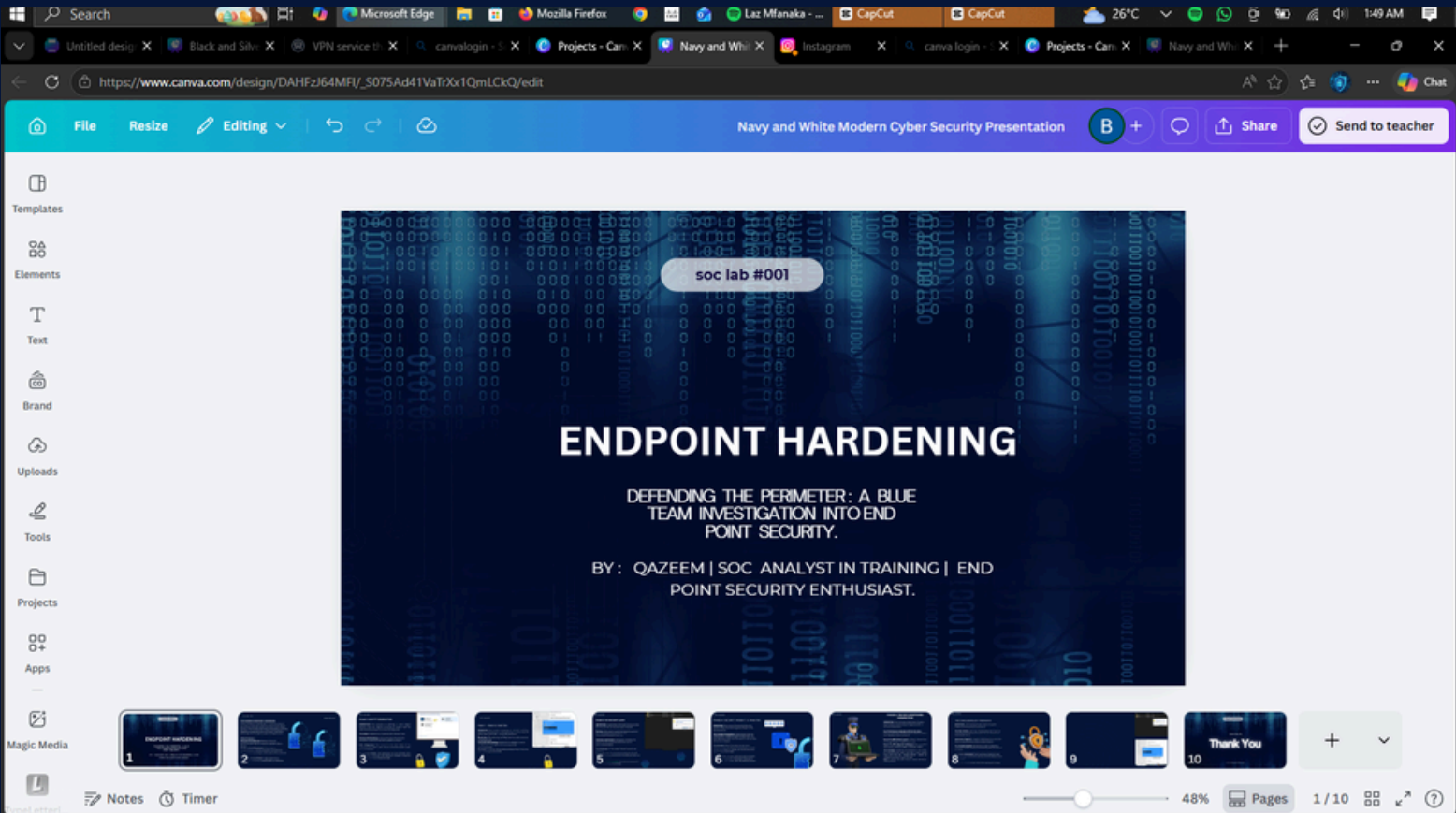
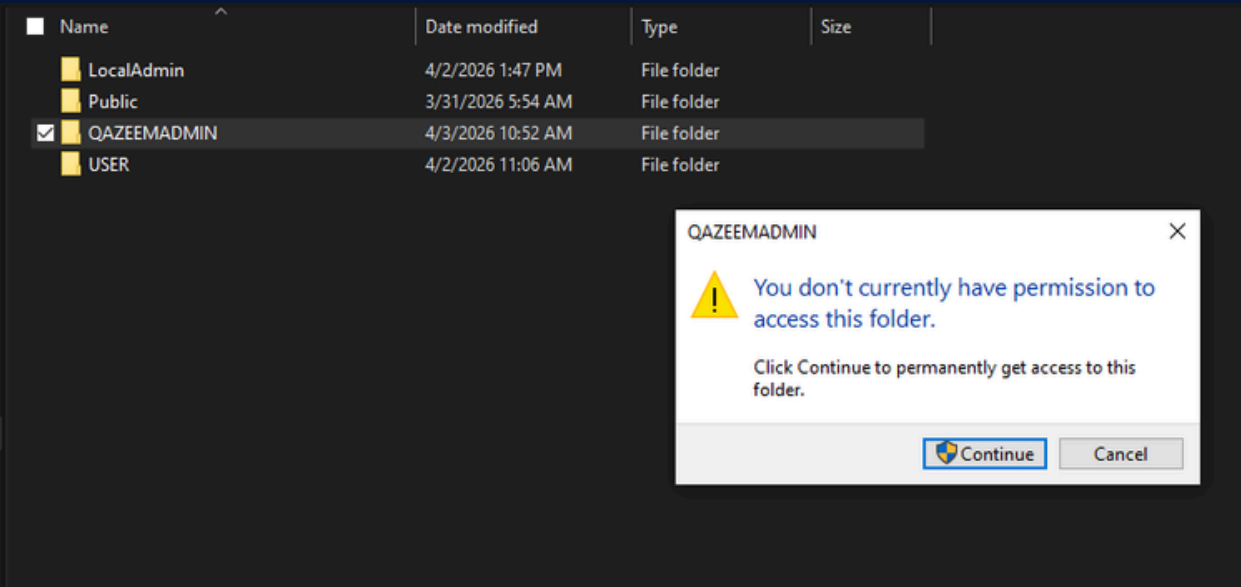
The Environment: This lab was conducted on a live Windows 11 endpoint to simulate real-world workstation hardening.

Technical Stack: Primary OS: Windows 10/11 Home/Pro.

Documentation: Created and structured in Canva to meet professional SOC reporting standards.

Evidence Capture: Captured via Windows Snipping Tool to ensure forensic accuracy.

The Process: From initial identity provisioning to final security auditing, every step was performed manually to verify the integrity of the Principle of Least Privilege.



Choose the user you would like to change



QAZEEM SOC
Local Account
Password protected



LocalAdmin
Local Account
Administrator
Password protected

SOC LAB #001

THANK YOU FOR YOUR TIME

Let's Secure the Future Together.

I am a Criminology & Security Studies student dedicated to bridging the gap between investigative logic and technical endpoint defense. I am currently seeking opportunities in SOC Operations, IT Support, and Digital Forensics.

Let's Connect:

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Portfolio: SOC Lab Series #001

The best defense is a proactive mindset. Let's start the conversation.